INCH−POUND

MIL−DTL−51222D(EA)
22 October 1997
SUPERSEDING
MIL−F−51222C(EA)
29 March 1988

DETAIL SPECIFICATION

FILTER, GAS, 150 CFM, M23A1

Inactive for new design after 28 September 1995

This specification is approved for use by the Edgewood, Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type and one size of toxic gas filter.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD−ENE−S, Aberdeen Proving Ground, MD 21010−5423 by using the self−addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL−F−46032 − Fungus−Resistant Treatment for Sandbags; Copper Processes

STANDARDS

FEDERAL

FED−STD−191 − Textile Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111−5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY EDGEWOOD RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

PURCHASE DESCRIPTIONS

EA−C−1704 − Carbon, Activated, Impregnated, Copper−Silver−Zinc−Molybdenum − Triethylenediamine (ASZM − TEDA)

DRAWINGS

5−19−2298 − Filter, Gas, 150 CFM, M23A1 (ASZM−TEDA)
E136−41−5000 − Tester, Filter Life, Q262, Assembly

(Copies are available from Commander, U.S. Army Information Management, ATTN: SIO-RI−ITD, Rock Island, IL 61299−5000.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

2.3 Non−Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

ASTM STANDARDS

D 2867 − Moisture in Activated Carbon

(Application for copies should be addressed to ASTM, 100 Barr Harbor Drive, West Conshohocken, PA 19428−2959.)

AIR−CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

ARI STANDARD 700 − Standard for Specifications for Fluorocarbons and other Refrigerants

(Application for copies should be addressed to ARI, 4301 North Fairfax Drive, Arlington, VA 22203.)

(Non−Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing 5−19−2298 and subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings. Cotton cloth shall be made mildew resistant in accordance with MIL−F−46032, using type III, method optional process except that leaching shall be omitted. No visible growth of mildew shall be evident when tested in accordance with method 5750 of FED−STD−191. Cotton cloth shall be water repellent and have a minimum spray rating of 50.0 when tested in accordance with method 5526 of FED−STD−191.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

3.2 Assembly. The assembly of the gas filter shall be consistent with Drawing 5−19−2298 (see 6.3).

3.3 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection in accordance with 4.2.

3.4 Air flow resistance. The air flow resistance of the filter shall not exceed 3.8 inches (97 mm) of water at a 150 standard cubic feet per minute (scfm) air flow rate with an air stream temperature of 21°C (70°F) and a barometric pressure of one atmosphere (760 mm Hg) when tested as specified in 4.3.6.1.

3.5 Filter Leakage. The filter shall not leak when a concentration of 1000 parts per million (ppm) of R−134a (1,1,1,2 Tetrafluoroethane) refrigerant is introduced at the inlet of the filter at a flow rate of 30 scfm when tested in accordance with 4.3.6.2. The refrigerant (R−134a) shall conform to the characteristics of ARI STANDARD 700. A filter leak shall be defined as the detection of 1 ppm or more of R134a in the effluent air within 2 minutes after the introduction of refrigerant at the inlet of the filter.

3.6 Moisture content of gas filter. The moisture content of the carbon in each filter immediately before packaging shall not exceed 3.0 percent by weight when tested as specified in 4.3.6.3.

3.7 Rough handling. After roughing handling in accordance with 4.3.6.4, sample filters shall show no evidence of cracks, dents, or carbon granule leakage upon visual inspection.

3.8 Dimethylmethylphosphonate (DMMP) life. The filter shall have a DMMP gas life of no less than 55 minutes at rated flow when tested in accordance with 4.3.6.5.

3.9 Cyanogen Chloride (CK) life. Each sample carbon obtained from the filling process of the filter shall have a minimum CK life of no less than 50 minutes when tested in tubes in accordance with 4.3.6.6.

3.10 Identification and markings. The marking of each filter shall include the name of the item (Filter, Gas 150 CFM, M23A1) and lot number as a minimum. Verification shall be in accordance with 4.3.5. Additional information may be required by solicitation (see 6.2).

3.11 Workmanship. The filter shall be free from foreign matter (e.g. dirt, grease, oil) and damage such as chipped, burred or bent sections, or loose, broken or frayed gasket material. Verification shall be in accordance with 4.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Quality conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Sample. The first article sample shall consist of 12 consecutively produced filters which pass airflow and leakage requirements (3.4 and 3.5). The first article sample shall be manufactured and packaged using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.2.2 Inspection procedure. As determined by the Government, the sample first article items shall be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions.

4.2.2.1 For examination. The sample packaged filters shall be examined for all requirements of the applicable drawings and this specification.

4.2.2.2 For test.

(a) Carbon testing.

(1) Samples for moisture testing. A sample of carbon shall be taken during the filling of the 1st, 6th and 12th filter. Each carbon sample taken during the filling of those 3 filters shall weigh at least 100 grams. Each sample shall be placed in an open container whose diameter is such that the depth of the carbon will be 1.5±0.5 inches deep. Those samples of carbon shall be exposed to the same conditions of temperature and humidity as their associated filters and be used for subsequent moisture testing. Concurrent with or immediately following the packaging of the filters corresponding to each carbon sample, each of three carbon samples shall be tested for moisture content in accordance with 4.3.6.3. If any carbon sample fails to meet the moisture requirement, the first article lot of filters represented by the samples shall be rejected.

(2) Samples for CK gas life testing. A sample of carbon shall be taken during the filling of 8 of the 12 first article filters, placed in an open container, and exposed to the same conditions of temperature and humidity as its associated filter. Each carbon sample taken during the filling of those filters shall weigh at least 100 grams. The 8 samples shall then be sealed in their containers and sent to the Government for subsequent CK life testing. The Government will test

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

8 carbon subsamples for CK life in accordance with 4.3.6.6. If any sample fails to meet the requirements for CK life, the first article lot represented shall be rejected.

(b) Filter testing. Each filter shall be tested for air flow resistance (4.3.6.1) and leakage (4.3.6.2). Following leakage testing, filters shall be packaged using the same methods, materials, equipment, and processes as will be used during regular production. After moisture analysis of the carbon samples (see 4.2.2.2(a)), the packaged sample filters which pass the moisture requirement in 3.6 shall be forwarded to the Government and subjected to rough handling in accordance with 4.3.6.4. Following rough handling, the Government will test ten (10) sample filters for DMMP life in accordance with 4.3.6.5. If any sample from the lot fails to meet the requirements for rough handling and DMMP life, the lot represented shall be rejected.

4.2.3 Acceptance criteria. If any first article sample filter fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.3 Quality conformance inspection.

4.3.1 Lotting. A lot shall consist of the filters produced by one manufacturer on one production line, from the same materials, with the same process, and without a break in production of more than 10 calendar days. However, any one lot of filters shall contain no more than one lot of carbon, and a new lot shall be started whenever any carbon filling or final assembly process equipment change is made. Each lot shall be identified by an alphanumeric lot number. The lot number shall include a manufacturer's identification symbol consisting of 3 alpha characters, a numeric code identifying the year of production, a code or abbreviation that signifies the month of production, and an interfix−serial number. The interfix−serial number shall change if there is change in the design, manufacturing process, materials, suppliers, production run, or if a new contract is issued.

4.3.2 Sampling. Sampling of packaged filters shall be conducted in accordance with the classification of characteristics in 4.3.5. Samples shall be selected at random. Carbon sample size for moisture testing and CK life testing shall be in accordance with 4.3.3.2. Sample size for rough handling in 4.3.6.4 shall be the same as that for DMMP life test in 4.3.6.5.

4.3.3 Inspection procedure.

4.3.3.1 For examination. Sample filters shall be inspected according to the classification of characteristics table in 4.3.5. If a non−conforming filter to a 100% inspected characteristic is found, the non−conforming filter shall be rejected and removed from the lot. Failure of any sample filter to conform to any critical or major characteristic in the classification of characteristics table (see 4.3.5) that is not covered by 100% inspection shall be cause for rejection of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

lot represented. Failure of any sample item to conform to any minor characteristic may be cause for rejection of the lot represented.

4.3.3.2 For test.

(a) Carbon testing.

(1) Samples for moisture testing. A sample of carbon shall be taken during the filling of the first and last filter manufactured each day. Each carbon sample taken during the filling of those 2 filters shall weigh at least 100 grams. Each sample shall be placed in an open container whose diameter is such that the depth of the carbon will be 1.5±0.5 inches deep. Those samples of carbon shall be exposed to the same conditions of temperature and humidity as their associated filters and be used for subsequent moisture testing. Concurrent with or immediately following the packaging of the filters corresponding to each carbon sample, each of the carbon samples shall be tested for moisture content in accordance with 4.3.6.3. If any of the carbon samples taken that day fails to meet the moisture requirement in 3.6, the production run of filters fabricated that day shall be rejected.

(2) Samples for CK life testing. For each lot, eight (8) samples of carbon, each sample of carbon weighing at least 100 grams, shall be taken at random times during the filling of 8 respective filters and placed in an open container. Those samples shall be exposed to the same conditions of temperature and humidity as their associated filters. The samples shall then be sealed in their containers and sent to the Government for subsequent CK life testing. The Government will determine the minimum CK life for 4 of the 8 samples for each lot in accordance with 4.3.6.6. If CK life is at least 57 minutes for each of four carbon samples, accept the lot; otherwise, perform CK testing on the remaining 4 carbon samples to accumulate CK life data on all 8 samples from the same lot. If CK life is less than 50 minutes for any of the 8 samples, reject the lot represented.

(b) Filter testing. Each filter shall be tested for air flow resistance (4.3.6.1) and leakage (4.3.6.2). Any filter failing air flow resistance or leakage requirements shall be rejected and removed from the lot, and then replaced, if possible (see 4.3.1), by one from the same lot which passes such requirements. Following leakage testing, filters shall be packaged using the same methods, materials, equipment, and processes as will be used during regular production. After moisture analysis of the carbon samples (see 4.3.3.2(a)), the packaged sample filters which pass the moisture requirement in 3.6 shall be forwarded to the Government and subjected to rough handling in accordance with 4.3.6.4. Following rough handling, sample filters shall be tested by the Government for DMMP life in accordance with 4.3.6.5. If any sample from the lot fails to meet the requirements for rough handling and DMMP life, the lot represented shall be rejected.

4.3.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.3.5 Classification of characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table I for selection of sample sizes.

TABLE I. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	**	**	**	**	**	**	**	**	**	**	**
9 to 15	**	**	**	**	**	**	**	**	**	**	**
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 and above	1250	315	192	116	86	68	50	38	29	22	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

**Uneconomical lot size. Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.5	Filter, gas, 150 cfm, M23A1	SHEET 1 OF 1	5−19−2298

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	Leakage	100 percent inspection	3.5	4.3.6.2
Major				
101	Components evident	Table I, level VIII	3.1	VI
102	Overall dimensions correct	Table I, level VIII	3.1	CE
103	Components correctly assembled	Table I, level VIII	3.2	VI
104	Airflow resistance	100 percent inspection	3.4	4.3.6.1
105	Moisture in carbon	See 4.3.3.2(a)(1)	3.6	4.3.6.3
106	Rough handling/DMMP life	Table I, level IX	3.8	VI & 4.3.6.5,
107	CK life	See 4.3.3.2(a)(2)	3.9	4.3.6.6
108	Workmanship	Table I, level VIII	3.11	VI
Minor				
201	Identification and marking correct	Table I, level IX	3.10	VI
202	Protective finish evident	Table I, level IX	3.1	VI

NOTES:

CE − Commercial inspection equipment
VI − Visual inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

4.3.6 Tests. Filters shall be tested in the following sequence:

4.3.6.1 Air flow resistance. The pressure drop across the filter shall be determined at the rated air flow of 150 scfm. The upstream and down−stream static pressure measuring tubes shall be as close as possible to the filter and shall not be on a section of duct that has a changing cross sectional area. Test for air flow resistance as follows:

(a) Connect the filter to a source of forced air.
(b) Set the flow of air through the filter to 150 scfm
. (c) Record the barometric pressure.
(d) Measure and record the air stream temperature.
(e) Determine and record the difference in static pressure head up−stream of the filter to that down−stream of the filter. NOTE: If barometric pressure and air stream temperature are not within 760±8 mm and 70±2°F, calculate the correction factor below to correct the measured pressure drop across filter to standard conditions (1 atm and 70°F). If barometric pressure and air stream temperature are within 760±8 mm and 70±2°F, a correction factor of unity may be assumed.

$$\Delta P(cal) = \Delta P(measured) \times CF, \text{ where}$$

$$CF = \frac{P(test) \times 86.2}{(460 + T)^{1.768}}$$

Notes:

CF − Correction factor
ΔP(cal) − Air flow resistance corrected to standard conditions in inches of water gage (iwg)
P(test) − Barometric pressure at time of test in mm Hg
ΔP(measured) − pressure drop recorded from test measurement (iwg)
T − Temperature of air stream flowing through filter in degrees F

4.3.6.2 Leakage. Rigidly position the filter to be tested with the long axis in the vertical position. Connect the filter to a source of forced air and place a mixing chamber at the influent side of the filter. Regulate the flow of air from the blower to 1/5 the rated airflow of the filter (30 scfm). Introduce R−134a into the intake of the blower; monitor and maintain a concentration of 1000 ppm by volume of R−134a in airflow of 30 scfm on the influent side of the filter at 75 ± 10° F and relative humidity (RH) no higher than 50 percent. Leakage shall be determined using a suitable leak detector (see 6.7) for sampling and detecting R−134a at the effluent side of the filter. Purge the filter by passing fresh air in the reverse direction of that indicated above through the filter at 150 scfm for 1 minute. Keep the exposure of the filter to air to a minimum. The airflow time for leakage should not exceed 3 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

4.3.6.3 Moisture content of gas filter. After mixing the carbon sample weighing at least 100 grams in the open container, obtain a 5 to 10 gm carbon subsample for the moisture content test. Determine the moisture content of each carbon subsample in accordance with ASTM D 2867, Oven−Drying Test Method, except that the oven temperature shall be 217°−225°F (103°−107°C) and the drying time shall be 3 hours. If the moisture content of the carbon exceeds the requirement in 3.6, the filters represented by the subsamples may be considered unacceptable (see note 6.4).

4.3.6.4 Rough handling. Position and clamp the unpackaged filter on a steel movable plate such that the influent side is facing up. The plate shall measure 30 by 30 inches and have a total weight of 240 ± 20 pounds including the holding clamps. When the filter is securely clamped in place, there shall be no distortion of the frame or body of the filter. The steel base plate shall be part of the apparatus and shall be firmly anchored to a concrete floor capable of absorbing the impact of the vertical vibrations. The steel movable plate and filter shall be raised and let fall (free) 3/4 ± 1/8 inch onto a steel base plate at the rate of 200 drops per minute for 15 ± 0.1 minutes. The test shall be performed at a room temperature of 70 ± 25 °F. Throughout the test the filter shall remain in a firm position. Filters may be padded or blocked where clamps press against the side or top of the frame, but not the bottom, to prevent distortion of the filter. The movable steel plate must be parallel to the base plate at all times. Samples shall be examined for compliance with 3.7.

4.3.6.5 DMMP life. After rough handling in accordance with 4.3.6.4, DMMP life shall be determined by the Government laboratory using the Q262 tester (Drawing E136−41−5000). Subject the filter to a DMMP challenge of 5± 0.25 milligrams per liter at its rated flow of 150 scfm. During the challenge, the temperature of the filter and air stream shall be 120±5 °F (49±3 °C). The DMMP life shall be determined for a break (endpoint) concentration of 0.04 micrograms per liter downstream of filter.

4.3.6.6 CK life. After mixing each sample carbon obtained during the filling of each filter, perform CK life testing on a carbon subsample placed in tubes in accordance with procedures, conditions and parameters in EA−C−1704.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the man-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

aging Military Department's or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gas filter covered by this specification is intended to be used in air purifiers that are part of gas−particulate filter units in military vehicles such as the Multiple Launch Rocket System.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see section 2)
(c) First article:
(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.
(d) Packaging requirements. See Special Packaging Instruction P5−19−2298.
(e) Additional identification or marking requirements, if required, must be in the contract and cite the identification method and responsibility for the items affected, such as test units, components, serial numbers, Julian date, lot numbers, contract number, etc.

6.3 Manufacture (see 3.2). The following method of manufacture has been found to produce a satisfactory gas filter and is provided for information only. The Government makes no warranty that material made in accordance with this method will pass the specification requirements.

(a) A filter frame with the bottom performated sheet, fines filter and drawbolt assemblies should be placed in a suitable gravity type filling machine.

(b) A measured amount of carbon conforming to EA−C−1704 should be dumped rapidly in the hopper of a specially designed gravity filling machine. The carbon should be swept over the surfaces of the distributing funnels. The funnel outlets shall be of a size that will not let the sorbent fall into the filter frame in less than 1−1/2 minutes. Six inches below the funnel outlets, a series of three 4 mesh screens are spaced two inches apart for spreading the sorbent as it falls into the filter frame. The funnels are spaced so that an excess of carbon is deposited near the edge of the filter frame for redistribution into the aluminum channel later. The distance between the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

funnel outlets and filter frame should be 36 inches. It is advisable to predetermine experimentally the required amount of charcoal to fill a frame.

(c) After being filled, the filter frame should be placed on a table and the carbon packed beneath the aluminum channel flanges with the fingers or a suitable tool. The carbon should be carefully brushed off the frame onto the sorbent bed and the bed leveled with a leveling device which directs the carbon away from the frame and levels the bed evenly 1/4−inch above the frame.

(d) The upper perforated sheet with attached fines filter should be placed onto the drawbolts, aligned properly with the edge of the frame and secured with several self−tapping screws. The four composition gaskets, steel washers and nuts should be assembled and the nuts hand tightened.

(e) The frame should be placed on a suitable vibrating press and the perforated sorbent retainer pressed firmly down on the frame, applying a compressed air pressure of 40 pounds per square inch (psi) which acting through a 6 inch diameter air cylinder exerts a force of about 1100 pounds on the frame. The self−tapping screws should then be assembled all around the periphery of the perforated retainer. The nuts on the drawbolts should be tightened to approximately 40 inch pounds. The filter should be vibrated for 30 seconds at a frequency of 200 strokes per minute and amplitude of one−half inch using an air pressure of 5 psi on the compressed air cylinder (this corresponds to a pressure on the frame of about 140 pounds.) After 30 seconds, the vibrator should be stopped and the drawbolt nuts retightened to 40 inch pounds. The vibration procedure should be repeated for an additional 30 seconds, and at the end of that time, the drawbolt nuts should be retightened to 40 inch pounds. After retightening the four bolts, the thickness of the filter should be calipered in the area immediately adjacent to each bolt. The thickness of the filter should not be less than 2.03 inches at these four points. The filter should be removed from the press, the drawbolts should be cut off flush with the nuts, and the nuts and washers coated with nonhardening Pernatex 2.

(f) The perforated sheet in the areas immediately adjacent to each end of the bolts (four) should be examined. If the perforated sheet is torn or separated, the filter should be rejected and reworked. The difficulty may be insufficient carbon or overtorquing.

6.4 Moisture control. One method to control moisture content in carbon to 3 percent or less is to control the humidity in the air during the carbon filling and filter assembly process. Any daily production runs of filters whose average moisture content exceeds 3 percent may be dried by passing dry air at less than 3% relative humidity through the filters for 20 hours at room temperature.

6.5 DMMP and CK life tests. Samples for DMMP and CK tests shall be forwarded by the contractor, together with DD Form 1222, to Director, U.S. Army Edgewood, Research, Devel-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

opment and Engineering Center, ATTN: SCBRD−ENM−N, Building E5165, Aberdeen Proving Ground, MD 21010−5423.

6.6 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.7 Leakage detector. The halide leak detector, model F1000, manufactured by Nuclear Consulting Services, Inc. has been found suitable for this purpose. When using this detector, the presence of high vapor pressure halogenated contaminants in the refrigerant could interfere with the filter leak test, resulting in false indications of filter leakage. Should false leakage indications be suspected, the refrigerant should be checked for the presence of such contaminants.

6.8 Lot size. A lot size of less than 50 should be avoided if possible to prevent more than 10 percent of the lot from being destroyed during DMMP life testing.

6.9 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject term (key word) listing.

Air purifier
CK (cyanogen chloride)
DMMP
Filtered air
Gas filter
R−134a (1,1,1,2 tetrafluoroethane)
Toxic gas filter

Custodian:

Army − EA

Preparing activity:

Army − EA

Project No. 4240−A240

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-51222D	**2. DOCUMENT DATE (YYMMDD)** 971022

3. DOCUMENT TITLE FILTER, GAS, 150 CFM, M23A1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME TECHNICAL DIRECTOR, U.S. ARMY EDGEWOOD RESEARCH, DEVELOPMENT AND ENG CTR	b. TELEPHONE *Include Area Code)* (1) Commercial (410) 671-1879 (2) AUTOVON 584-1879
c. ADDRESS *(Include Zip Code)* ATTN: SCBRD-ENE-S ABERDEEN PROVING GROUND, MD 21010-5423	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:53Z
Check the source to verify that this is the current version before use.